

Neurons and Bone mechanics

1. Suppose that the temporal activity of the space-clamped nerve axon is modelled with the two variable FitzHugh-Nagumo model with an external applied current I_a :

$$\frac{dv}{dt} = f(v) - w + I_a, \quad \frac{dw}{dt} = bv - w,$$

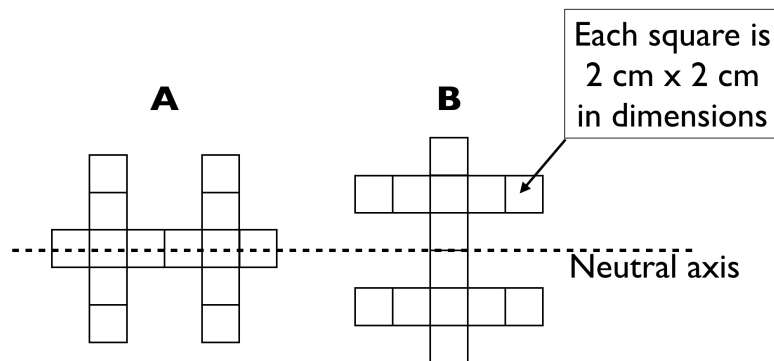
where the function $f(v)$ is the piecewise linear function:

$$f(v) = \begin{cases} -v, & \text{for } v \leq 1 \\ \frac{1}{2}v - \frac{3}{2}, & \text{for } 1 \leq v \leq 5 \\ 6 - v, & \text{for } v \geq 5 \end{cases}$$

and I_a is a positive constant.

Performing an analysis of nullclines, find the range of values of the parameters b and I_a for which the model could have a periodic behaviour.

2. Consider two beams A and B of equal cross-sectional area, with the material distributed around the neutral axis/plane as shown in the figure below. Note that the area of the square used to construct the shapes is equal to $2\text{cm} \times 2\text{cm} = 4\text{cm}^2$. Calculate the area moment of inertia I_A for these two cases. The beam of which shape, A or B , will resist the bending better? Why?



1. $\frac{dv}{dt} = 0, w = I_a + f(v)$

$\frac{dw}{dt} = 0, w = bv$

the model have a periodic behaviour:

$$\begin{cases} b > \frac{1}{2} \\ b < I_a + 1 \\ 5b > I_a + 1 \end{cases}$$

$$\therefore \begin{cases} b > \frac{1}{2} \\ b + 1 < I_a < 5b - 1 \end{cases}$$

2.

$$\begin{aligned} A: \quad I_A &= 4 \int_{-1}^1 y^2 \cdot 2 dy + 2 \int_{-5}^5 y^2 \cdot 2 dy \\ &= \frac{1016}{3} \text{ cm}^4 \end{aligned}$$

$$\begin{aligned} B: \quad I_B &= \int_{-6}^{-4} y^2 \cdot 2 dy + \int_4^6 y^2 \cdot 2 dy + \int_{-1}^2 y^2 \cdot 2 dy \\ &\quad + \int_{-2}^2 y^2 \cdot 10 dy + \int_2^4 y^2 \cdot 10 dy \\ &= \frac{1760}{3} \text{ cm}^4 \end{aligned}$$

$$I_A < I_B$$

I_B resist the bending better.